

CS 6384 Computer Vision

Assignment 2

Due: Friday, March 6

Download the [homework2_programming.zip](#) file from eLearning, Assignment 2. Finish the following programming problems and submit your scripts to eLearning. You can zip all the data and files for submission. We will run your scripts to verify them.

Install the Python packages needed by

- `pip install -r requirement.txt`

Here are some useful resources:

- Python basics <https://pythonbasics.org/>
- Numpy <https://numpy.org/doc/stable/user/basics.html>
- OpenCV https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html

1 Problem 1

(5 points) Harris Corner Detector.

Implement the `harris_corner()` function and the `non_maximum_suppression()` function in `harris_corner.py`. The script implements the Harris corner detection algorithm. Follow the steps in the script to implement it. After your implementation, run the `harris_corner.py` in Python to verify it. Figure 1 shows an example of running the script.

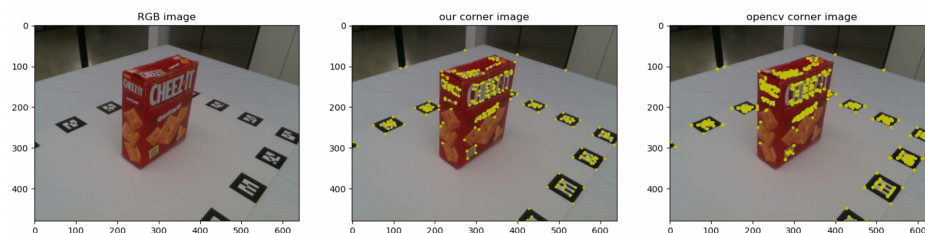


Figure 1: (Left) An input image. (Middle) Harris corner detection with our implementation. (Right) Harris corner detection using the provided function in OpenCV. The yellow dots indicate the detected corners.

2 Problem 2

(5 points) SIFT feature matching.

Implement the `sift_matching()` function in `sift_matching.py`. This script first extracts the SIFT keypoints and descriptors from two images with OpenCV. Then it calls the `sift_matching()` function to match the detected SIFT keypoints from the two images. After your implementation, run the `sift_matching.py` in Python to verify it. Figure 2 shows an example of running the script.

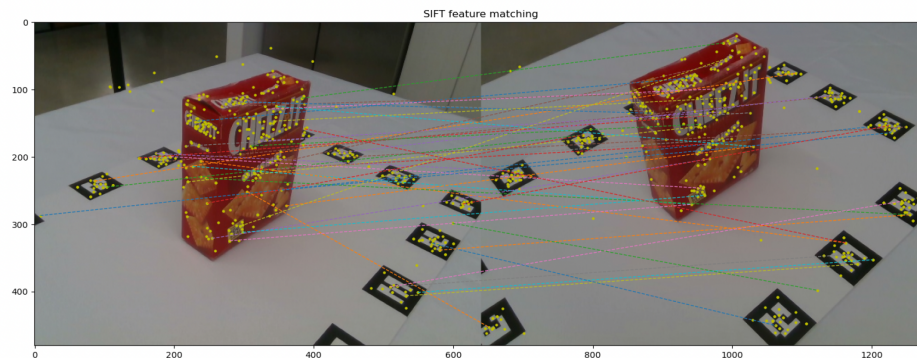


Figure 2: SIFT feature matching between two images. The yellow dots indicate the detected SIFT keypoints and the lines show the matching.